Postal Address: PO Box 881, Emalahleni, 1035
Physical Address: 6 Newton Street, Industrial Area, Ext 11, Middelburg, 1055
Tel: 010 170 8304

CERTIFICATE OF ANALYSIS

Client Name	eMalahleni Local Municipality
Contact number and e-mail	013 690 6300/S311569@Emalahleni.gov.za
Date Reported	28.01.2026

C-202600012

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Disclaimer: Results represent sample(s) as received. The analytical results do not apply to any similar item that has not been analysed. Neither Mfulawamanzi nor any of its officials shall be liable for damages arising from loss or injury caused directly or indirectly by or contributed to by or arising from the results produced or the interpretation thereof. This report shall not be reproduced except in full and provided that the approval of the Management of Mfulawamanzi has been secured.

Note: Bacteriological samples cannot be retained. Re-sampling will have to be done should any queries relating to the results occur.

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CERTIFICATE OF ANALYSIS

C-202600012

ilac-MRA sanas T0911

Subject	**Water Analyses**	**Client Name**	**eMalahleni Local Municipality**	**Contract / Order No.**	**ELM-43/2023**
Water Source	**Potable**	**Client Physical Address**	**Private Bag x 3**	**Contract Date:**	**01,11,2024**
Date of Collection	**19.01.2026**		**Witbank , Mpumalanga , 1035**	**Expiry Date:**	**01,11,2027**
Date of Receipt	**19.01.2026**	**Contact person**	**Thembani Vukeya**	**SABS No.**	**B112**
Date Analysis Started	**20.01.2026**	**Contact number and e-mail**	**013 690 6300/ S311569@Emalahleni.gov.za**	**NLA No.**	**cvgrd**
Date Completed	**26.01.2026**	**Sampler/Courier**	**Zinhle Themba**	**DWA No.**	**163**

ANALYSIS REQUIRED	Sample Name	Witbank Raw	Witbank Final	Uncertainty (95%)	SANS 241:2025 STANDARD LIMITS	Method Number
	LAB REF. NO	ELMP-26/0098	ELMP-26/0099			
	Condition of Sample	Brownish with particles	Clear			
PHYSICAL AND AESTHETIC DETERMINANDS	**UNIT**					
pH at 25˚C	pH units	7.04	7.19	0.0465	≥5 to ≤ 9.7	TM-03
Turbidity	NTU	6.81	1.26	0.0657	≤ 1	TM-05
Electrical Conductivity at 25˚C	mS/m	75.60	78.20	0.0243	≤ 170	TM-04
**True Colour	mg/L Pt-Co	13.0	12.0	-	≤ 15	TM-21
CHEMICAL DETERMINANDS - MACRO DETERMINANDS	**UNIT**					
Nitrate as N	mg/L	0.38	0.40	0.0687	≤ 11	TM-08
Nitrite as N	mg/L	0.001	0.013	0.0532	≤ 0.9	TM-07
**Combined Nitrate plus Nitrite as N	mg/L	0.034	0.020	-	<1	ns
**Sulphate as SO_4	mg/L	237.45	226.79	-	≤ 250	TM-40
**Free and Saline Ammonia as N	mg/L	0.054	0.030	0.0934	≤ 0.30	TM-13
	mg/L	0.054	0.030	0.0934	≤ 1.5	TM-13
**Fluoride as F	mg/L	<0.40	<0.40	0.0428	≤ 1.5	TM-22
**Chloride as Cl	mg/L	24	35	0.0693	≤ 250	TM-23
**Sodium as Na	mg/L	24.12	25.33	-	≤ 200	TM-33
CHEMICAL DETERMINANDS - MICRO DETERMINANDS	**UNIT**					
**Aluminium as Al	mg/L	0.052	0.086	-	≤ 0.200	TM-32
**Iron as Fe	mg/L	0.032	0.048	-	≤ 0.300	TM-32
**Lead as Pb	mg/L	0.003	0.003	-	≤ 0.010	TM-32
**Zinc as Zn	mg/L	0.003	0.019	-	≤ 3	TM-32
**Copper as Cu	mg/L	0.050	0.050	-	≤ 2.000	TM-32
**Manganese as Mn	mg/L	0.318	0.031	-	≤ 0.100	TM-32
ALERT OPERATIONAL	**UNIT**					
**m-Alkalinity as CaCO3	mg/L	107	108	ns	ns	TM-18
**Total Hardness as CaCO3	mg/L	302	316	ns	ns	TM-28
**Calcium Hardness as CaCO3	mg/L	146	146	ns	ns	TM-28
**Magnesium Hardness as CaCO3	mg/L	157	170	ns	ns	TM-28
**p Alkalinity	mg/L	0	0	ns	ns	TM-18
**Calcium as Ca	mg/L	58.35	58.71	ns	ns	TM-33
**Magnesium as Mg	mg/L	38.11	41.22	ns	ns	TM-33
**Potassuim as k	mg/L	6.11	6.32	ns	ns	TM-33
**Temperature	°C	25	25	-	ns	ns
CALCULATIONS						
**Langerlier Index	index	-1.00	-0.84	ns	ns	TM-30
Cation - Anions Balancing Calculations (Water Quality Analysis)						
****Corrositivity Index**	**index**	2.43	2.43	ns	ns	ns
****%Balancing**	**%**	97.13	98.61	ns	ns	ns
****TDS Calculated**	**mg/L**	456	466	ns	≤ 1200	TM-34

NOTES:

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CERTIFICATE OF ANALYSIS

C-202600012

ilac-MRA sanas T0911

Subject	**Water Analyses**	**Client Name**	**eMalahleni Local Municipality**	**Contract / Order No.**	**ELM-43/2023**
Water Source	**Potable**	**Client Physical Address**	**Private Bag x 3**	**Contract Date:**	**01,11,2024**
Date of Collection	**19.01.2026**		**Witbank , Mpumalanga , 1035**	**Expiry Date:**	**01,11,2027**
Date of Receipt	**19.01.2026**	**Contact person**	**Thembani Vukeya**	**SABS No.**	**B112**
Date Analysis Started	**20.01.2026**	**Contact number and e-mail**	**013 690 6300/ S311569@Emalahleni.gov.za**	**NLA No.**	**cvgrd**
Date Completed	**26.01.2026**	**Sampler/Courier**	**Zinhle Themba**	**DWA No.**	**163**

ANALYSIS REQUIRED	**Sample Name**	**Rietspruit Raw**	**Rietspruit Final**	**Uncertainty (95%)**	**SANS 241:2025 STANDARD LIMITS**	**Method Number**
	LAB REF. NO	**ELMP-26/0100**	**ELMP-26/0101**			
	Condition of Sample	**Brownish with particles**	**Clear**			
PHYSICAL AND AESTHETIC DETERMINANDS	**UNIT**					
pH at 25˚C	pH units	7.37	7.08	0.0465	≥5 to ≤ 9.7	TM-03
Turbidity	NTU	5.16	1.61	0.0657	≤ 1	TM-05
Electrical Conductivity at 25˚C	mS/m	45.50	48.60	0.0243	≤ 170	TM-04
**True Colour	mg/L Pt-Co	38.0	25.0	-	≤ 15	TM-21
CHEMICAL DETERMINANDS - MACRO DETERMINANDS	**UNIT**					
Nitrate as N	mg/L	0.18	0.20	0.0687	≤ 11	TM-08
Nitrite as N	mg/L	0.005	0.006	0.0532	≤ 0.9	TM-07
**Combined Nitrate plus Nitrite as N	mg/L	0.011	0.011	-	<1	ns
**Sulphate as SO_4	mg/L	75.28	69.06	-	≤ 250	TM-40
**Free and Saline Ammonia as N	mg/L	0.017	0.008	0.0934	≤ 0.30	TM-13
	mg/L	0.017	0.008	0.0934	≤ 1.5	TM-13
**Fluoride as F	mg/L	0.49	0.48	0.0428	≤ 1.5	TM-22
**Chloride as Cl	mg/L	33	49	0.0693	≤ 250	TM-23
**Sodium as Na	mg/L	25.82	26.14	-	≤ 200	TM-33
CHEMICAL DETERMINANDS - MICRO DETERMINANDS	**UNIT**					
**Aluminium as Al	mg/L	0.318	0.358	-	≤ 0.200	TM-32
**Iron as Fe	mg/L	0.247	0.016	-	≤ 0.300	TM-32
**Lead as Pb	mg/L	0.003	0.004	-	≤ 0.010	TM-32
**Zinc as Zn	mg/L	0.017	0.016	-	≤ 3	TM-32
**Copper as Cu	mg/L	0.050	0.050	-	≤ 2.000	TM-32
**Manganese as Mn	mg/L	0.126	0.055	-	≤ 0.100	TM-32
ALERT OPERATIONAL	**UNIT**					
**m-Alkalinity as CaCO3	mg/L	102	91	ns	ns	TM-18
**Total Hardness as CaCO3	mg/L	138	156	ns	ns	TM-28
**Calcium Hardness as CaCO3	mg/L	75	97	ns	ns	TM-28
**Magnesium Hardness as CaCO3	mg/L	63	59	ns	ns	TM-28
**p Alkalinity	mg/L	0	0	ns	ns	TM-18
**Calcium as Ca	mg/L	29.86	38.77	ns	ns	TM-33
**Magnesium as Mg	mg/L	15.33	14.31	ns	ns	TM-33
**Potassuim as k	mg/L	7.41	7.26	ns	ns	TM-33
**Temperature	°C	25	25	-	ns	ns
CALCULATIONS						
**Langerlier Index	index	-0.89	-1.10	ns	ns	TM-30
Cation - Anions Balancing Calculations (Water Quality Analysis)						
****Corrositivity Index**	**index**	1.07	1.30	ns	ns	ns
*****%Balancing**	**%**	97.87	98.22	ns	ns	ns
****TDS Calculated**	**mg/L**	253	265	ns	≤ 1200	TM-34

NOTES:

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CERTIFICATE OF ANALYSIS

C-202600012

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Subject	**Water Analyses**	**Client Name**	**eMalahleni Local Municipality**	**Contract / Order No.**	**ELM-43/2023**
Water Source	**Potable**	**Client Physical Address**	**Private Bag x 3**	**Contract Date:**	**01,11,2024**
Date of Collection	**19.01.2026**		**Witbank , Mpumalanga , 1035**	**Expiry Date:**	**01,11,2027**
Date of Receipt	**19.01.2026**	**Contact person**	**Thembani Vukeya**	**SABS No.**	**B112**
Date Analysis Started	**20.01.2026**	**Contact number and e-mail**	**013 690 6300/ S311569@Emalahleni.gov.za**	**NLA No.**	**cvgrd**
Date Completed	**26.01.2026**	**Sampler/Courier**	**Zinhle Themba**	**DWA No.**	**163**

ANALYSIS REQUIRED	Sample Name / LAB REF. NO / Condition of Sample	Kriel Raw ELMP-26/0102 Clear with particles	Kriel Final ELMP-26/0103 Clear	Uncertainty (95%)	SANS 241:2025 STANDARD LIMITS	Method Number
PHYSICAL AND AESTHETIC DETERMINANDS	**UNIT**					
pH at 25˚C	pH units	7.22	7.08	0.0465	≥5 to ≤ 9.7	TM-03
Turbidity	NTU	2.50	8.23	0.0657	≤ 1	TM-05
Electrical Conductivity at 25˚C	mS/m	<10	<10	0.0243	≤ 170	TM-04
**True Colour	mg/L Pt-Co	8.0	1.0	-	≤ 15	TM-21
CHEMICAL DETERMINANDS - MACRO DETERMINANDS	**UNIT**					
Nitrate as N	mg/L	0.17	0.17	0.0687	≤ 11	TM-08
Nitrite as N	mg/L	0.006	0.004	0.0532	≤ 0.9	TM-07
**Combined Nitrate plus Nitrite as N	mg/L	0.008	0.011	-	<1	ns
**Sulphate as SO_4	mg/L	13.79	14.65	-	≤ 250	TM-40
**Free and Saline Ammonia as N	mg/L	0.011	0.008	0.0934	≤ 0.30	TM-13
	mg/L	0.011	0.008	0.0934	≤ 1.5	TM-13
**Fluoride as F	mg/L	<0.40	<0.40	0.0428	≤ 1.5	TM-22
**Chloride as Cl	mg/L	<10	11	0.0693	≤ 250	TM-23
**Sodium as Na	mg/L	2.95	4.88	-	≤ 200	TM-33
CHEMICAL DETERMINANDS - MICRO DETERMINANDS	**UNIT**					
**Aluminium as Al	mg/L	0.058	0.868	-	≤ 0.200	TM-32
**Iron as Fe	mg/L	0.069	0.017	-	≤ 0.300	TM-32
**Lead as Pb	mg/L	0.005	0.003	-	≤ 0.010	TM-32
**Zinc as Zn	mg/L	0.019	0.019	-	≤ 3	TM-32
**Copper as Cu	mg/L	0.050	0.050	-	≤ 2.000	TM-32
**Manganese as Mn	mg/L	0.007	0.002	-	≤ 0.100	TM-32
ALERT OPERATIONAL	**UNIT**					
**m-Alkalinity as CaCO3	mg/L	22	23	ns	ns	TM-18
**Total Hardness as CaCO3	mg/L	27	34	ns	ns	TM-28
**Calcium Hardness as CaCO3	mg/L	14	17	ns	ns	TM-28
**Magnesium Hardness as CaCO3	mg/L	13	17	ns	ns	TM-28
**p Alkalinity	mg/L	0	0	ns	ns	TM-18
**Calcium as Ca	mg/L	5.60	6.82	ns	ns	TM-33
**Magnesium as Mg	mg/L	3.10	4.04	ns	ns	TM-33
**Potassuim as k	mg/L	2.08	2.07	ns	ns	TM-33
**Temperature	°C	25	25	-	ns	ns
CALCULATIONS						
**Langerlier Index	index	-2.20	-2.30	ns	ns	TM-30
Cation - Anions Balancing Calculations (Water Quality Analysis)						
****Corrositivity Index**	**index**	0.84	1.12	ns	ns	ns
****%Balancing**	**%**	98.73	98.24	ns	ns	ns
****TDS Calculated**	**mg/L**	49	60	ns	≤ 1200	TM-34

NOTES:

Decision Rule (Pass/Fail) , Final Assessment:

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CERTIFICATE OF ANALYSIS

C-202600012

sanas T0911

Subject	**Water Analyses**	**Client Name**	**eMalahleni Local Municipality**	**Contract / Order No.**	**ELM-43/2023**
Water Source	**Potable**	**Client Physical Address**	**Private Bag x 3**	**Contract Date:**	**01,11,2024**
Date of Collection	**21.01.2026**		**Witbank , Mpumalanga , 1035**	**Expiry Date:**	**01,11,2027**
Date of Receipt	**21.01.2026**	**Contact person**	**Thembani Vukeya**	**SABS No.**	**B112**
Date Analysis Started	**22.01.2026**	**Contact number and e-mail**	**013 690 6300/ S311569@Emalahleni.gov.za**	**NLA No.**	**cvgrd**
Date Completed	**26.01.2026**	**Sampler/Courier**	**Zinhle Themba**	**DWA No.**	**163**

ANALYSIS REQUIRED	Sample Name / LAB REF. NO / Condition of Sample	Nu Water Raw Point S (Benfleur) / ELMP-26/0104 / Brownish	Nu Water Final Point S (Benfleur) / ELMP-26/0105 / Clear	Uncertainty (95%)	SANS 241:2025 STANDARD LIMITS	Method Number
PHYSICAL AND AESTHETIC DETERMINANDS	**UNIT**					
pH at 25˚C	pH units	7.24	9.04	0.0465	≥5 to ≤ 9.7	TM-03
Turbidity	NTU	4.17	0.609	0.0657	≤ 1	TM-05
Electrical Conductivity at 25˚C	mS/m	76.00	81.70	0.0243	≤ 170	TM-04
**True Colour	mg/L Pt-Co	12.0	11.0	-	≤ 15	TM-21
CHEMICAL DETERMINANDS - MACRO DETERMINANDS	**UNIT**					
Nitrate as N	mg/L	0.293	0.33	0.0687	≤ 11	TM-08
Nitrite as N	mg/L	0.012	0.005	0.0532	≤ 0.9	TM-07
**Combined Nitrate plus Nitrite as N	mg/L	0.042	0.024	-	<1	ns
**Sulphate as SO_4	mg/L	242.33	249.91	-	≤ 250	TM-40
**Free and Saline Ammonia as N	mg/L	0.089	0.011	0.0934	≤ 0.30	TM-13
	mg/L	0.089	0.011	0.0934	≤ 1.5	TM-13
**Fluoride as F	mg/L	<0.40	<0.40	0.0428	≤ 1.5	TM-22
**Chloride as Cl	mg/L	23	32	0.0693	≤ 250	TM-23
**Sodium as Na	mg/L	32.08	48.38	-	≤ 200	TM-33
CHEMICAL DETERMINANDS - MICRO DETERMINANDS	**UNIT**					
**Aluminium as Al	mg/L	0.075	0.025	-	≤ 0.200	TM-32
**Iron as Fe	mg/L	0.065	0.092	-	≤ 0.300	TM-32
**Lead as Pb	mg/L	0.002	0.002	-	≤ 0.010	TM-32
**Zinc as Zn	mg/L	0.019	0.002	-	≤ 3	TM-32
**Copper as Cu	mg/L	0.050	0.050	-	≤ 2.000	TM-32
**Manganese as Mn	mg/L	0.082	0.166	-	≤ 0.100	TM-32
ALERT OPERATIONAL	**UNIT**					
**m-Alkalinity as CaCO3	mg/L	105	134	ns	ns	TM-18
**Total Hardness as CaCO3	mg/L	195	209	ns	ns	TM-28
**Calcium Hardness as CaCO3	mg/L	99	105	ns	ns	TM-28
**Magnesium Hardness as CaCO3	mg/L	96	104	ns	ns	TM-28
**p Alkalinity	mg/L	0	22	ns	ns	TM-18
**Calcium as Ca	mg/L	39.84	42.10	ns	ns	TM-33
**Magnesium as Mg	mg/L	23.28	25.22	ns	ns	TM-33
**Potassuim as k	mg/L	8.23	8.13	ns	ns	TM-33
**Temperature	°C	25	25	-	ns	ns
CALCULATIONS						
**Langerlier Index	index	-0.97	0.95	ns	ns	TM-30
Cation - Anions Balancing Calculations (Water Quality Analysis)						
****Corrositivity Index**	**index**	2.52	2.11	ns	ns	ns
****%Balancing**	**%**	83.69	86.01	ns	ns	ns
****TDS Calculated**	**mg/L**	440	491	ns	≤ 1200	TM-34

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CERTIFICATE OF ANALYSIS

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sanas T0911

Subject	**Water Analyses**	**Client Name**	**eMalahleni Local Municipality**	**Contract / Order No.**	**ELM-43/2023**
Water Source	**Potable**	**Client Physical Address**	**Private Bag x 3**	**Contract Date:**	**01,11,2024**
Date of Collection	**19.01.2026**		**Witbank , Mpumalanga , 1035**	**Expiry Date:**	**01,11,2027**
Date of Receipt	**19.01.2026**	**Contact person**	**Thembani Vukeya**	**SABS No.**	**B112**
Date Analysis Started	**20.01.2026**	**Contact number and e-mail**	**013 690 6300/ S311569@Emalahleni.gov.za**	**NLA No.**	**cvgrd**
Date Completed	**26.01.2026**	**Sampler/Courier**	**Zinhle Themba**	**DWA No.**	**163**

ANALYSIS REQUIRED	Sample Name / LAB REF. NO / Condition of Sample	Nu Water Raw Point E ELMP-26/0106 Brownish	Nu Water Final Point E ELMP-26/0107 Clear	Uncertainty (95%)	SANS 241:2025 STANDARD LIMITS	Method Number
PHYSICAL AND AESTHETIC DETERMINANDS	**UNIT**					
pH at 25˚C	pH units	7.77	7.54	0.0465	≥5 to ≤ 9.7	TM-03
Turbidity	NTU	2.17	0.148	0.0657	≤ 1	TM-05
Electrical Conductivity at 25˚C	mS/m	76.00	79.30	0.0243	≤ 170	TM-04
**True Colour	mg/L Pt-Co	14.0	7.0	-	≤ 15	TM-21
CHEMICAL DETERMINANDS - MACRO DETERMINANDS	**UNIT**					
Nitrate as N	mg/L	0.47	0.46	0.0687	≤ 11	TM-08
Nitrite as N	mg/L	0.009	0.009	0.0532	≤ 0.9	TM-07
**Combined Nitrate plus Nitrite as N	mg/L	0.032	0.031	-	<1	ns
**Sulphate as SO_4	mg/L	234.77	234.68	-	≤ 250	TM-40
**Free and Saline Ammonia as N	mg/L	0.022	0.013	0.0934	≤ 0.30	TM-13
	mg/L	0.022	0.013	0.0934	≤ 1.5	TM-13
**Fluoride as F	mg/L	<0.40	<0.40	0.0428	≤ 1.5	TM-22
**Chloride as Cl	mg/L	24	33	0.0693	≤ 250	TM-23
**Sodium as Na	mg/L	24.93	29.43	-	≤ 200	TM-33
CHEMICAL DETERMINANDS - MICRO DETERMINANDS	**UNIT**					
**Aluminium as Al	mg/L	0.029	0.041	-	≤ 0.200	TM-32
**Iron as Fe	mg/L	0.004	0.048	-	≤ 0.300	TM-32
**Lead as Pb	mg/L	0.003	0.003	-	≤ 0.010	TM-32
**Zinc as Zn	mg/L	0.001	0.008	-	≤ 3	TM-32
**Copper as Cu	mg/L	0.050	0.050	-	≤ 2.000	TM-32
**Manganese as Mn	mg/L	0.058	0.050	-	≤ 0.100	TM-32
ALERT OPERATIONAL	**UNIT**					
**m-Alkalinity as CaCO3	mg/L	103	110	ns	ns	TM-18
**Total Hardness as CaCO3	mg/L	299	310	ns	ns	TM-28
**Calcium Hardness as CaCO3	mg/L	148	155	ns	ns	TM-28
**Magnesium Hardness as CaCO3	mg/L	151	155	ns	ns	TM-28
**p Alkalinity	mg/L	0	0	ns	ns	TM-18
**Calcium as Ca	mg/L	59.45	62.30	ns	ns	TM-33
**Magnesium as Mg	mg/L	36.69	37.61	ns	ns	TM-33
**Potassuim as k	mg/L	6.52	6.30	ns	ns	TM-33
**Temperature	°C	25	25	-	ns	ns
CALCULATIONS						
**Langerlier Index	index	-0.27	-0.46	ns	ns	TM-30
Cation - Anions Balancing Calculations (Water Quality Analysis)						
****Corrositivity Index**	**index**	2.50	2.43	ns	ns	ns
*****%Balancing**	**%**	97.61	98.07	ns	ns	ns
****TDS Calculated**	**mg/L**	455	476	ns	≤ 1200	TM-34

NOTES:

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3. Results that are shaded and bold means they didn't meet specified standard limits

Dtini

Dunga Edmond Ntini - Technical Signatory

CERTIFICATE OF ANALYSIS

C-202600012

T0911

Postal Address: PO Box 881, Emalahleni, 1035
Physical Address: 6 Newton Street, Industrial Area, Ext 11, Middelburg, 1055
Tel: 010 170 8304

Subject	**Water Analyses**	**Client Name**	**eMalahleni Local Municipality**	**Contract / Order No.**	**ELM-43/2023**
Water Source	**Potable**	**Client Physical Address**	**Private Bag x 3**	**Contract Date:**	**01,11,2024**
Date of Collection	**19.01.2026**		**Witbank , Mpumalanga , 1035**	**Expiry Date:**	**01,11,2027**
Date of Receipt	**19.01.2026**	**Contact person**	**Thembani Vukeya**	**SABS No.**	**B112**
Date Analysis Started	**20.01.2026**	**Contact number and e-mail**	**013 690 6300/ S311569@Emalahleni.gov.za**	**NLA No.**	**cvgrd**
Date Completed	**26.01.2026**	**Sampler/Courier**	**Zinhle Themba**	**DWA No.**	**163**

ANALYSIS REQUIRED	**Sample Name**	**Hlalanikahle Bulk Water Augumentation Plant Raw**	**Hlalanikahle Bulk Water Augumentation Plant Final**	**Uncertainty (95%)**	**SANS 241:2025 STANDARD LIMITS**	**Method Number**
	LAB REF. NO	**ELMP-26/0108**	**ELMP-26/0109**			
	Condition of Sample	**Clear**	**Clear**			
PHYSICAL AND AESTHETIC DETERMINANDS	**UNIT**					
pH at 25˚C	pH units	4.76	4.65	0.0465	≥5 to ≤ 9.7	TM-03
Turbidity	NTU	0.185	0.242	0.0657	≤ 1	TM-05
Electrical Conductivity at 25˚C	mS/m	23.40	23.40	0.0243	≤ 170	TM-04
**True Colour	mg/L Pt-Co	1.0	1.0	-	≤ 15	TM-21
CHEMICAL DETERMINANDS - MACRO DETERMINANDS	**UNIT**					
Nitrate as N	mg/L	11.44	11.52	0.0687	≤ 11	TM-08
Nitrite as N	mg/L	0.006	0.005	0.0532	≤ 0.9	TM-07
**Combined Nitrate plus Nitrite as N	mg/L	1.033	1.041	-	<1	ns
**Sulphate as SO_4	mg/L	21.46	21.91	-	≤ 250	TM-40
**Free and Saline Ammonia as N	mg/L	0.003	0.003	0.0934	≤ 0.30	TM-13
	mg/L	0.003	0.003	0.0934	≤ 1.5	TM-13
**Fluoride as F	mg/L	<0.40	<0.40	0.0428	≤ 1.5	TM-22
**Chloride as Cl	mg/L	25	25	0.0693	≤ 250	TM-23
**Sodium as Na	mg/L	22.16	25.15	-	≤ 200	TM-33
CHEMICAL DETERMINANDS - MICRO DETERMINANDS	**UNIT**					
**Aluminium as Al	mg/L	0.204	0.218	-	≤ 0.200	TM-32
**Iron as Fe	mg/L	0.048	0.048	-	≤ 0.300	TM-32
**Lead as Pb	mg/L	0.003	0.002	-	≤ 0.010	TM-32
**Zinc as Zn	mg/L	0.020	0.017	-	≤ 3	TM-32
**Copper as Cu	mg/L	0.050	0.050	-	≤ 2.000	TM-32
**Manganese as Mn	mg/L	0.104	0.104	-	≤ 0.100	TM-32
ALERT OPERATIONAL	**UNIT**					
**m-Alkalinity as CaCO3	mg/L	4	2	ns	ns	TM-18
**Total Hardness as CaCO3	mg/L	40	34	ns	ns	TM-28
**Calcium Hardness as CaCO3	mg/L	21	20	ns	ns	TM-28
**Magnesium Hardness as CaCO3	mg/L	20	14	ns	ns	TM-28
**p Alkalinity	mg/L	0	0	ns	ns	TM-18
**Calcium as Ca	mg/L	8.39	7.92	ns	ns	TM-33
**Magnesium as Mg	mg/L	4.74	3.52	ns	ns	TM-33
**Potassuim as k	mg/L	2.38	2.30	ns	ns	TM-33
**Temperature	°C	25	25	-	ns	ns
CALCULATIONS						
**Langerlier Index	index	-5.40	-5.80	ns	ns	TM-30
Cation - Anions Balancing Calculations (Water Quality Analysis)						
****Corrositivity Index**	**index**	12.79	23.06	ns	ns	ns
****%Balancing**	**%**	97.73	98.25	ns	ns	ns
****TDS Calculated**	**mg/L**	140	141	ns	≤ 1200	TM-34

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CERTIFICATE OF ANALYSIS

C-202600012

T0911

Subject	**Water Analyses**	**Client Name**	**eMalahleni Local Municipality**	**Contract / Order No.**	**ELM-43/2023**
Water Source	**Potable**	**Client Physical Address**	**Private Bag x 3**	**Contract Date:**	**01,11,2024**
Date of Collection	**19.01.2026**		**Witbank , Mpumalanga , 1035**	**Expiry Date:**	**01,11,2027**
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Date Analysis Started	**20.01.2026**	**Contact number and e-mail**	**013 690 6300/ S311569@Emalahleni.gov.za**	**NLA No.**	**cvgrd**
Date Completed	**26.01.2026**	**Sampler/Courier**	**Zinhle Themba**	**DWA No.**	**163**

ANALYSIS REQUIRED	**Sample Name**	**Nu Water Raw Point B**	**Nu Water Final Point B**	**Uncertainty (95%)**	**SANS 241:2025 STANDARD LIMITS**	**Method Number**
	LAB REF. NO	**ELMP-26/0110**	**ELMP-26/0111**			
	Condition of Sample	**Brownish with particles**	**Clear**			
PHYSICAL AND AESTHETIC DETERMINANDS	**UNIT**					
pH at 25˚C	pH units	7.27	7.77	0.0465	≥5 to ≤ 9.7	TM-03
Turbidity	NTU	4.97	0.644	0.0657	≤ 1	TM-05
Electrical Conductivity at 25˚C	mS/m	75.80	79.30	0.0243	≤ 170	TM-04
**True Colour	mg/L Pt-Co	14.0	8.0	-	≤ 15	TM-21
CHEMICAL DETERMINANDS - MACRO DETERMINANDS	**UNIT**					
Nitrate as N	mg/L	0.37	0.38	0.0687	≤ 11	TM-08
Nitrite as N	mg/L	0.001	0.003	0.0532	≤ 0.9	TM-07
**Combined Nitrate plus Nitrite as N	mg/L	0.035	0.031	-	<1	ns
**Sulphate as SO_4	mg/L	241.968	235.14	-	≤ 250	TM-40
**Free and Saline Ammonia as N	mg/L	0.052	0.007	0.0934	≤ 0.30	TM-13
	mg/L	0.052	0.007	0.0934	≤ 1.5	TM-13
**Fluoride as F	mg/L	<0.40	<0.40	0.0428	≤ 1.5	TM-22
**Chloride as Cl	mg/L	24	31	0.0693	≤ 250	TM-23
**Sodium as Na	mg/L	24.60	29.55	-	≤ 200	TM-33
CHEMICAL DETERMINANDS - MICRO DETERMINANDS	**UNIT**					
**Aluminium as Al	mg/L	0.074	0.041	-	≤ 0.200	TM-32
**Iron as Fe	mg/L	0.011	0.002	-	≤ 0.300	TM-32
**Lead as Pb	mg/L	0.004	0.002	-	≤ 0.010	TM-32
**Zinc as Zn	mg/L	0.002	0.019	-	≤ 3	TM-32
**Copper as Cu	mg/L	0.050	0.050	-	≤ 2.000	TM-32
**Manganese as Mn	mg/L	0.231	0.047	-	≤ 0.100	TM-32
ALERT OPERATIONAL	**UNIT**	.				
**m-Alkalinity as CaCO3	mg/L	104	113	ns	ns	TM-18
**Total Hardness as CaCO3	mg/L	312	312	ns	ns	TM-28
**Calcium Hardness as CaCO3	mg/L	159	161	ns	ns	TM-28
**Magnesium Hardness as CaCO3	mg/L	152	151	ns	ns	TM-28
**p Alkalinity	mg/L	0	3.96	ns	ns	TM-18
**Calcium as Ca	mg/L	63.79	64.62	ns	ns	TM-33
**Magnesium as Mg	mg/L	37.02	36.64	ns	ns	TM-33
**Potassuim as k	mg/L	6.49	6.36	ns	ns	TM-33
**Temperature	°C	25	25	-	ns	ns
CALCULATIONS						
**Langerlier Index	index	-0.74	-0.21	ns	ns	TM-30
Cation - Anions Balancing Calculations (Water Quality Analysis)						
****Corrositivity Index**	**index**	2.56	2.35	ns	ns	ns
*****%Balancing**	**%**	98.22	97.99	ns	ns	ns
****TDS Calculated**	**mg/L**	464	476	ns	≤ 1200	TM-34

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